STAGE 0

Dataset row

tasks/videomme/utils.py

--tasks · picks _w_subtitle, _long, _no_visual

data_files · 200-row local parquet vs. 2,700 upstream

--limit, --offset,

test_split, HF_HOME

→ doc + /path/<id>.mp4

STAGE 1

Temporal sampling

qwen_vl_utils:smart_nframes

nframes = 32 · fixed count, evenly spaced over the whole clip

fps ! · mutually exclusive: setting it drops nframes

FRAME_FACTOR = 2 ! · counts

round down to even

→ N x H0 x W0 x 3

STAGE 2

Spatial resize

qwen_vl_utils:smart_resize

max_pixels = 301 056 ! · hard-clamped at 602 112 for video

min_pixels = 200 704 · smaller frames are upscaled

sides snap to multiples of 28

→ N x H x W (mod 28)

STAGE 3

Serialization

protocol.py:to_openai_msgs

pass_video_url = False ! · N <|image_pad|> blocks, not one <|video_pad|>

not reachable from --model vllm

LMMS_IMAGE_ENCODE_FORMAT

= PNG

→ N x base64 image_url

STAGE 4

Prompt construction

doc_to_text + kwargs

post_prompt · "Answer with the option's letter … directly."

subtitle preamble ! · names subtitles even when none are passed

profiles keyed by the literal

--model string

→ messages[N imgs, text]

STAGE 5

Context budget

chat template + vLLM engine

max_model_len = 32768 · too small truncates silently

limit_mm_per_prompt ! · {"video":1} caps a modality that never reaches the engine

gpu_memory_utilization

= 0.8

→ ≈6144 vision tok

STAGE 6

Generation

simple/vllm.py:sampling

max_new_tokens ! · max(task 16, model 4096) is a *floor*, not a cap

temperature = 0, top_p = 0.95

num_beams, do_sample are

dropped in silence

→ "A" or essay ⊃ "A"

STAGE 7

Parse & score

process_results + metrics

--reasoning_tags · <think> stripped before matching

extraction regex is per task · bare letter vs. mcq_acc

cluster_key = videoID

widens the SE, correctly

→ acc = 0.6150 ± SE

! **Knobs that do not do what their name suggests.** fps and nframes are mutually exclusive and never merged · max_frame_num is read only on the fps branch · odd frame counts round down (ask 7, get 6) · max_pixels is silently clamped to 602 112 for video frames · pass_video_url cannot be reached from --model vllm · max_new_tokens is a floor, so --gen_kwargs alone will not lower it.

Σ **Stages 1–2 fix the context bill paid at stage 5.** vision_tokens ≈ (frames/2) × (max_pixels/784), where the /2 is the temporal patch merge and $784 = 28 \times 28$ pixels per token after the 2×2 spatial merge; 32 frames at max_pixels = 301 056 is ≈6,144 vision tokens for a single multiple-choice question.